

Foundation (Jalapeno)

- Q1.** What is the purpose of a scatter plot in space exploration?
- (A) To display planetary alignments
 - (B) To show the relationship between two variables
 - (C) To illustrate the life cycle of stars
 - (D) To predict asteroid trajectories
 - (E) To visualize black holes
- Q2.** If a scatter plot shows a positive correlation, what can be inferred?
- (A) As one variable increases, the other decreases
 - (B) As one variable increases, the other also tends to increase
 - (C) There is no relationship between the variables
 - (D) The data points are randomly distributed
 - (E) The variables are unrelated to space exploration
- Q3.** What is the equation of a line of best fit for the data points (2, 3), (4, 5), and (6, 7)?
- (A) $y = x - 1$
 - (B) $y = x + 1$
 - (C) $y = 2x - 1$
 - (D) $y = 2x + 1$
 - (E) $y = x^2 + 1$
- Q4.** A scientist collects data on the temperature of a planet and its distance from the sun. If the scatter plot shows a negative correlation, what can be concluded?
- (A) As the distance from the sun increases, the temperature also tends to increase
 - (B) As the distance from the sun increases, the temperature tends to decrease
 - (C) There is no relationship between temperature and distance from the sun
 - (D) The data points are randomly distributed
 - (E) The temperature is unrelated to the distance from the sun
- Q5.** What is the purpose of drawing a line of best fit on a scatter plot?
- (A) To identify outliers in the data
 - (B) To visualize the relationship between two variables
 - (C) To make predictions based on data trends
 - (D) To calculate the mean and median of the data
 - (E) To determine the standard deviation of the data
- Q6.** A space agency collects data on the altitude of a spacecraft and its velocity. If the scatter plot shows a strong positive correlation, what can be inferred?
- (A) As the altitude increases, the velocity tends to decrease
 - (B) As the altitude increases, the velocity tends to increase
 - (C) There is no relationship between altitude and velocity
 - (D) The data points are randomly distributed
 - (E) The velocity is unrelated to the altitude
- Q7.** What is the equation of a line of best fit for the data points (1, 2), (3, 4), and (5, 6)?
- (A) $y = x - 1$
 - (B) $y = x + 1$
 - (C) $y = 2x - 1$
 - (D) $y = 2x + 1$
 - (E) $y = x^2 + 1$
- Q8.** A researcher collects data on the mass of a planet and its orbital period. If the scatter plot shows a weak positive correlation, what can be concluded?
- (A) As the mass increases, the orbital period tends to decrease
 - (B) As the mass increases, the orbital period tends to increase
 - (C) There is no relationship between mass and orbital period
 - (D) The data points are randomly distributed
 - (E) The mass is unrelated to the orbital period

Open Ended Questions (FRQ)

- Q9.** A space agency collects data on the distance of a spacecraft from Earth and its communication delay. The data points are (1000, 2), (2000, 4), and (3000, 6). Draw a scatter plot and find the equation of the line of best fit. Use the equation to predict the communication delay when the spacecraft is 4000 km away from Earth.
- Q10.** A scientist collects data on the temperature of a planet and its atmospheric pressure. The data points are (10, 20), (15, 30), and (20, 40). Draw a scatter plot and identify the type of correlation. Use the data to make a prediction about the atmospheric pressure when the temperature is 25 degrees.

Chef's Tips

- Tip 1: Encourage students to use graph paper to create scatter plots
- Tip 2: Remind students to consider the context of the problem when making predictions
- Tip 3: Emphasize the importance of labeling axes and including a title on the scatter plot